

SPS CODING

```
#include <HX711.h>

#include <Wire.h>

#include <LiquidCrystal_I2C.h>

// HX711 Load Sensor

#define DT 4

#define SCK 5

HX711 scale;

// RPM Sensor

#define RPM_PIN 14

volatile int rpmCount = 0;

// Hall Sensor

#define HALL_PIN 27

// Fuel Sensor (Analog)

#define FUEL_PIN 34

// Ultrasonic Sensor

#define TRIG 12

#define ECHO 13

// GSM

#define GSM_TX 17

#define GSM_RX 16

// LCD

LiquidCrystal_I2C lcd(0x27, 16, 2);

// Variables

float weight;

float fuelLevel;

long duration;
```

```
float distance;

int rpm;

void IRAM_ATTR rpmISR() {
  rpmCount++;
}

void setup() {
  Serial.begin(115200);

  // Load Sensor
  scale.begin(DT, SCK);
  scale.set_scale();
  scale.tare();

  // RPM
  pinMode(RPM_PIN, INPUT);
  attachInterrupt(digitalPinToInterrupt(RPM_PIN), rpmISR, RISING);

  // Hall
  pinMode(HALL_PIN, INPUT);

  // Ultrasonic
  pinMode(TRIG, OUTPUT);
  pinMode(ECHO, INPUT);

  / LCD
  lcd.init();
  lcd.backlight();

  // GSM
  Serial2.begin(9600, SERIAL_8N1, GSM_RX, GSM_TX);
  lcd.setCursor(0,0);
  lcd.print("System Ready");
  delay(2000);
```

```
}  
  
void loop() {  
  // Load Sensor  
  weight = scale.get_units(5);  
  
  // RPM Calculation  
  rpm = rpmCount * 60;  
  rpmCount = 0;  
  
  // Fuel Level  
  fuelLevel = analogRead(FUEL_PIN);  
  fuelLevel = map(fuelLevel, 0, 4095, 0, 100);  
  
  // Ultrasonic  
  digitalWrite(TRIG, LOW);  
  delayMicroseconds(2);  
  digitalWrite(TRIG, HIGH);  
  delayMicroseconds(10);  
  digitalWrite(TRIG, LOW);  
  duration = pulseIn(ECHO, HIGH);  
  distance = duration * 0.034 / 2;  
  
  // LCD DISPLAY  
  lcd.clear();  
  lcd.setCursor(0,0);  
  lcd.print("Load:");  
  lcd.print(weight);  
  lcd.setCursor(0,1);  
  lcd.print("Fuel:");  
  lcd.print(fuelLevel);  
  lcd.print("%");
```

```
delay(2000);

lcd.clear();

lcd.setCursor(0,0);

lcd.print("RPM:");

lcd.print(rpm);

lcd.setCursor(0,1);

lcd.print("Dist:");

lcd.print(distance);

delay(2000);

// ALERT CONDITIONS

if(weight > 70) {

sendSMS("High Soil Resistance! Reduce Speed");

}

if(distance < 20) {

sendSMS("Obstacle Detected! Stop Tractor");

}

if(fuelLevel < 20) {

sendSMS("Low Fuel Level! Refill Needed");

}

delay(3000);

}

// GSM FUNCTION

void sendSMS(String message) {

Serial2.println("AT+CMGF=1");

delay(1000);

Serial2.println("AT+CMGS=\"+91XXXXXXXXXX\""); // Replace number

delay(1000);
```

```
Serial2.println(message);  
delay(100);  
Serial2.write(26);  
delay(3000);  
}
```